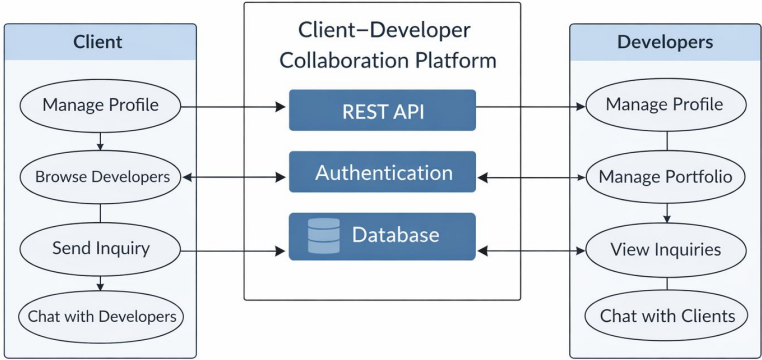




## Post Graduate Diploma in Advanced Computing (PG-DAC)

### August 2025 Batch

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| Group Number  | Group - 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Guide         | Mr.Pankaj Kumar Mahto                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Group Members | <ol style="list-style-type: none"> <li>1. Abuzer Amaan</li> <li>2. Kumar Apoorb</li> <li>3. Akshat Shokeen</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Title         | <b>Client and Developers Meets</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Objective     | <ol style="list-style-type: none"> <li>1. To develop an online platform where clients and developers can connect, communicate, and collaborate efficiently for project development.</li> <li>2. To allow clients to post their project requirements clearly and enable developers to understand project scope, budget, and deadlines easily.</li> <li>3. To provide developers a feature to create and manage their profiles and showcase their skills, technologies, and previous projects.</li> <li>4. To implement a secure login and registration system with role-based access control for clients and developers.</li> <li>5. To enable direct communication between clients and developers through messaging or chat functionality.</li> <li>6. To protect user data through password encryption, secure APIs, and proper database management.</li> </ol> |
| Abstract      | <p>The Client–Developer Meet System is a web-based platform designed to connect clients who require software solutions with developers who provide development services. The primary goal of the system is to create a structured and secure environment where clients can share project requirements and developers can showcase their skills and portfolios.</p> <p>The system provides role-based authentication for clients and developers, ensuring secure access and proper authorization. Clients can browse developer profiles, send project inquiries, and communicate directly through an integrated messaging system. Developers can manage their profiles, upload portfolio details, respond to inquiries, and interact with clients efficiently.</p>                                                                                                |

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| Project Architecture |  <pre> graph LR     subgraph Client         direction TB         C1([Manage Profile]) --&gt; C2([Browse Developers])         C2 --&gt; C3([Send Inquiry])         C3 --&gt; C4([Chat with Developers])     end      subgraph Platform [Client-Developer Collaboration Platform]         direction TB         P1[REST API]         P2[Authentication]         P3[(Database)]     end      subgraph Developers         direction TB         D1([Manage Profile]) --&gt; D2([Manage Portfolio])         D2 --&gt; D3([View Inquiries])         D3 --&gt; D4([Chat with Clients])     end      C1 --&gt; P1     C2 --&gt; P2     C3 --&gt; P3     P1 --&gt; D1     P2 --&gt; D2     P3 --&gt; D3 </pre> <p>The diagram illustrates the project architecture. It features three main components: a Client, a central Client-Developer Collaboration Platform, and Developers. The Client side includes a vertical flow of actions: Manage Profile, Browse Developers, Send Inquiry, and Chat with Developers. The Developers side includes: Manage Profile, Manage Portfolio, View Inquiries, and Chat with Clients. The central platform acts as a bridge, containing a REST API, Authentication module, and a Database. Bidirectional arrows connect the Client's 'Manage Profile', 'Browse Developers', and 'Send Inquiry' actions to the platform's REST API, Authentication, and Database respectively. Similarly, bidirectional arrows connect the Developers' 'Manage Profile', 'Manage Portfolio', and 'View Inquiries' actions to the platform's REST API, Authentication, and Database respectively.</p> |
| Scope of work        | <ol style="list-style-type: none"> <li>1. Develop a secure platform for client and developer registration with role-based authentication.</li> <li>2. Enable clients to post project requirements and send inquiries to developers.</li> <li>3. Allow developers to create and manage profiles and showcase portfolios.</li> <li>4. Provide a messaging/chat system for direct communication between clients and developers.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Technologies used    | <p>Programming Language: Java 21</p> <p>Front End: HTML, CSS, Thymeleaf</p> <p>Back End: Spring Boot 3.2.5</p> <p>Web Framework: Spring Boot Starter Web (Spring MVC, Embedded Tomcat)</p> <p>ORM Framework: Spring Data JPA (Hibernate)</p> <p>Validation: Spring Boot Starter Validation</p> <p>Database Driver: MySQL Connector/J</p> <p>Database: MySQL</p> <p>Build Tool: Maven</p> <p>IDE: Visual Studio Code (VS Code)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| <p>Application</p>                             | <p>It provides a user-friendly interface that allows clients and developers to easily register, log in, and manage their profiles.</p> <p>Updating profile information, portfolio details, and project inquiries becomes easier through the system.</p> <p>It gives appropriate access to authorized users depending on their roles (Client or Developer).</p> <p>The system ensures secure communication and data management using role-based authentication.</p> <p>The system has sufficient scope for modification and future enhancements if necessary.</p> |
| <p>Project Timelines<br/>(Total:120 hours)</p> | <p>Time plan of the Project:</p> <p>Requirement Analysis &amp; Abstract Preparation</p> <p>System Design (ER Diagram &amp; Database Design)</p> <p>Application Development (Frontend + Backend Coding)</p> <p>Testing &amp; Debugging (Code Testing &amp; Validation)</p> <p>Documentation &amp; Project Report Preparation</p>                                                                                                                                                                                                                                  |